

A Home with Family Values

Oak Ridge National Laboratory and Tuskegee University strike a balance between energy efficiency and affordability.

by Sherry E. Livengood
and Robert Wendt

Create a house that provides a healthy indoor environment fit to breathe in, that is energy efficient, and that a family with an annual income of \$12,000 can afford. Responding to the growing concerns of U.S. scientists and health professionals, that was the goal of a team at Oak Ridge National Laboratory's (ORNL) Buildings Technology Center and Tuskegee University's College of Engineering, Architecture, and Physical Sciences. A growing number of children from low income families suffer from asthma (see "Asthma and Children"). Our team's goal was to develop a prototypical house that balanced the sometimes competing values of indoor air quality (IAQ), affordability, and energy efficiency.

The major IAQ issues that our prototype is designed to address are excessive moisture, which causes mold and fungus to proliferate; dust mites and other allergens; volatile organic compounds (VOCs); and poor ventilation and combustion problems that reduce oxygen levels and increase levels of carbon monoxide (CO) and carbon dioxide (CO₂).

We intended the house to be affordable for people earning close to minimum wage. We defined an affordable house as one that could be purchased by a family of up to four people with one wage earner making \$12,000 per year, financed through a traditional market-based mortgage. Since maintenance and utility costs are a major factor in affordability, the house was also designed to be energy efficient and as maintenance-free as possible. The result is a 768 ft² house located at Tuskegee's experimental farm in Alabama. The healthy house has two bedrooms, one bath, a kitchen, a living room, and two small utility areas for the water heater and HVAC equipment (see Figure 1).



Making It Possible

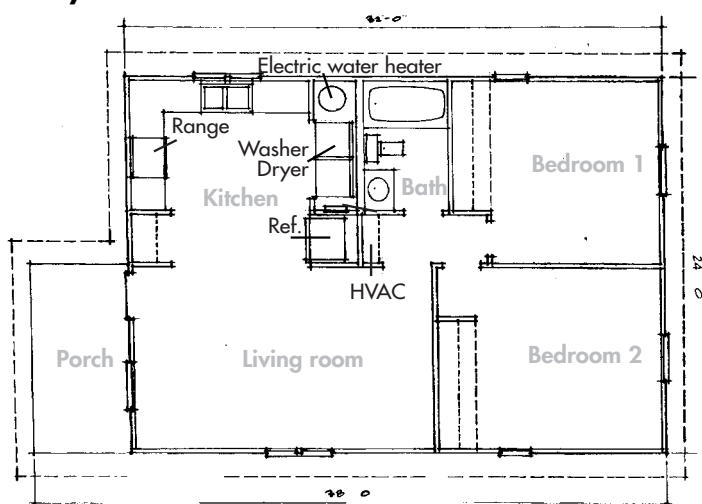
The design criteria for the ORNL-Tuskegee healthy house were adapted from the Clean Air Florida Home Standard and the American Lung Association Health House guidelines. (For more on the Health House guidelines, see "Home Performance for the Lungs," *HE* Mar/Apr '98, p. 9, and "Health House Cold Climate Standards Keep Occupants Warm and Dry," *HE* Nov/Dec '99, p. 34.) The ORNL-Tuskegee healthy house deviated from these standards only where they would compromise affordability and energy efficiency significantly. For example, a central vacuum cleaner was not installed because it would be expensive, and there is no carpet anywhere in the house. All the floors in the healthy house are made from a prefinished synthetic wood product that can be cleaned with a dust mop or a wet mop. Since a garage was considered an unnecessary luxury, there is no mud room (anteroom between the garage and the main living space), but wet shoes and outerwear can be placed on the covered porch to dry.

The bathroom does not have a window, but it does have a ventilation fan that is wired to the light switch. We considered installing a humidistat or time clock during the design process, but since every effort was being made to minimize equipment costs, and since a similar effect could be achieved by wiring the exhaust fan to the light switch, we decided to do that and keep it simple. This eliminated one more piece of equipment.

The house's insulated foam sheathing does have minimal outgassing, but this sheathing is installed outside the conditioned space and thus most outgassing would be to the outside. We decided that the energy efficiency benefit of the insulated sheathing outweighed the outgassing risks. In order to make the house more affordable for low-income consumers, the designers attempted to provide necessary features in as efficient a space as possible, thus minimizing the quantity and cost of materials.

We selected materials with an eye to cost control, low maintenance, energy efficiency, and air quality. Of these, our top priority was cost control. Sometimes

Healthy House Floor Plan



(left) **Figure 1.** The healthy house was built for a family of modest means. (above) Tuskegee professors Heshmat Aglang and Troy McQueen check out the healthy house during construction. Metal cross bracing is attached at each wall to prevent racking. Behind the bracing is a continuous layer of R-5 foam sheathing.

we had to make compromises. For example, we used vinyl siding even though it does outgas, because it is low-maintenance, inexpensive, and relatively durable. The siding is installed outside, where fresh air helps to dissipate the gases and odors.

Getting Down to Specifics

The concrete masonry unit (CMU) foundation wall has an unvented crawlspace with concrete footings under the exterior wall. The crawlspace floor was covered with polyurethane to prevent evaporation of moisture from the ground into the living area of the house. Initially the crawlspace will be uninsulated. After a period of monitoring, insulation will be installed in the crawlspace, and monitoring will continue. A central support wall of treated wood studs supports the floor joists and subfloor. The subfloor is made of 3/4-inch thick tongue-and-groove plywood panels. It is supported by 2 x 8 wood joists at 24 inches on center. Optimum Value Engineered Building System (OVE) 2 x 4 wood studs at 24 inches on center frame the exterior walls. Metal cross bracing is attached at each wall to prevent racking, and a two-stud corner is used to allow insulation to be placed at the corners of the exterior walls.

R-13 fiberglass batts are placed between the studs. The house is sheathed with R-5 insulated foam sheathing, installed outside the framing. This sheathing has minimal outgassing; it

should not significantly affect IAQ, but it adds a continuous layer of insulation as well as an air and moisture barrier. The framing and sheathing techniques were designed to minimize thermal shorts and improve energy efficiency. As mentioned above, vinyl siding was installed in order to lower the initial cost of construction and to minimize maintenance costs.

The roof support is made of prefabricated wood trusses that have raised heel joints at the exterior wall to allow roof insulation to extend to the edge of the framing. The roofing material selected was a relatively low-cost white metal roofing called 5-V Crimp. This roofing material has high reflectivity (see "Roofs Reflect Better Savings," *HE* July/Aug '01, p. 24) and durability, and it requires little maintenance over a 40- to 50-year life span. A continuous soffit and ridge vent system was installed to control the temperature in the attic. The attic has two layers of fiberglass insulation, for a total R-value of 30. One layer of insulation is placed between the bottom chords of the trusses and the next layer is laid perpendicular to the first layer. White vinyl double-pane windows and an insulated metal entrance door were also installed.

Interior partitions are constructed of steel studs with gypsum board walls that were finished and painted with low-VOC paint finish. A snap locking, prefinished, synthetic wood floor product was used on all the floors of the healthy house, with glued joints and caulked edges for wet locations. The

flooring was relatively expensive, but it required low skills to install and the benefit of having no carpet outweighed the cost of the materials. The kitchen and bathroom cabinets are solid wood and are coated with low-VOC finishes.

Conventional design and off-the-shelf materials were used for the electrical system. All penetrations through the building envelope were sealed for energy efficiency. Off-the-shelf wall-mounted light fixtures were installed to reduce penetrations through the ceiling and minimize stack effect exfiltration. We considered using CFLs but decided against it. The first cost for CFLs is higher, and this was an important factor to consider in keeping the house affordable. Every room except the bathroom has daylighting, which means that lights are needed, at most, a few hours a day. The electrical load from lights is only a small percentage of the total load for this residence.

The HVAC system is an optimally sized conventional split-package heat pump. Installing a heat pump eliminates potential combustion by-products. The house has a general ventilation system, with a 0.3-micron HEPA filter at the supply and spot exhaust units in the bath and kitchen. All appliances are electric, which also helps to eliminate combustion by-products. We considered using more sophisticated equipment, such as a heat recovery ventilator or a desuperheater heat exchanger for the air conditioning system, but we finally rejected these

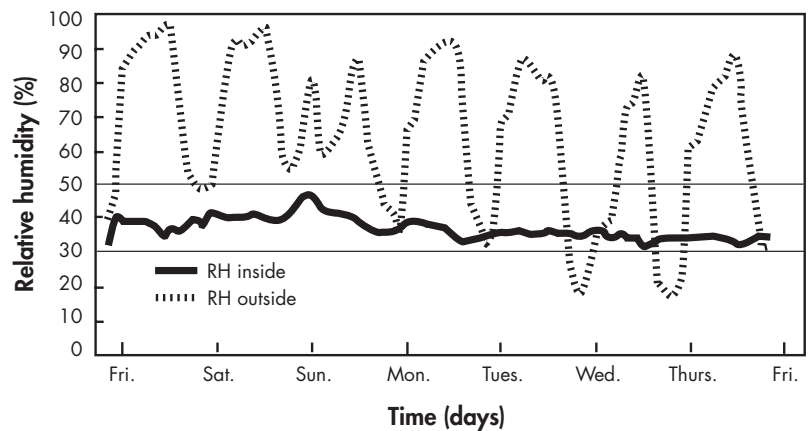
options because they are not cost-effective for an affordable house in rural Alabama. All ducts are within the conditioned space. The only penetration of the ceiling is the waste vent.

Monitoring Healthy Home Performance

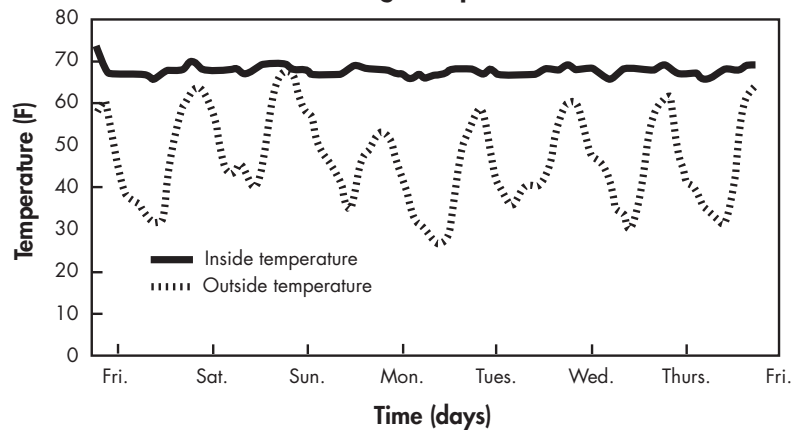
Now that construction has been completed, air quality and energy monitoring devices have been installed and we have gathered preliminary data. These monitoring devices will enable ORNL and Tuskegee staff to measure and assess the level of IAQ and energy efficiency under various conditions. Many published documents provide recommendations for constructing, renovating, or maintaining healthy IAQ, but for the most part, the recommendations are based on common knowledge or conjecture, not on measured data in prototypical structures. Furthermore, most of these documents do not factor cost or energy efficiency into the equation. Monitoring the house at Tuskegee will begin to provide reliable, measured data that can be used to improve the overall house as a system.

By experimenting with a variety of configurations, researchers will be able to determine how to optimize both IAQ and energy efficiency in a cost-effective manner. Sensors will monitor indoor and outdoor temperatures, indoor and outdoor relative humidity (RH), energy consumption, and ventilation fan time of operation, as well as CO₂, VOCs, and particulates. Radon will be spot monitored in the crawlspace. The house will be monitored first without occupants, then with various numbers of single-event occupants (party load), and finally with full-time residents.

First Week of Monitoring: RH



First Week of Monitoring: Temperature



Figures 2 and 3. The electrical meter reading at the end of the 168-hour test period was 84 kWh expended. The ventilation fan and every other electric device except the heat pump and the data logger were turned off.

Early Indications

During the first week of monitoring (February 8–15, 2002), measurements were taken with the heat pump on and without running the ventilation fan—there was no fresh ventilation air (see

Figures 2 and 3). CO₂ levels ranged from 380 to 460 ppm, and VOCs were measured at the optimal level (see “Green One”). The electrical meter reading at the end of the week was 84 kWh expended.

Asthma and Children

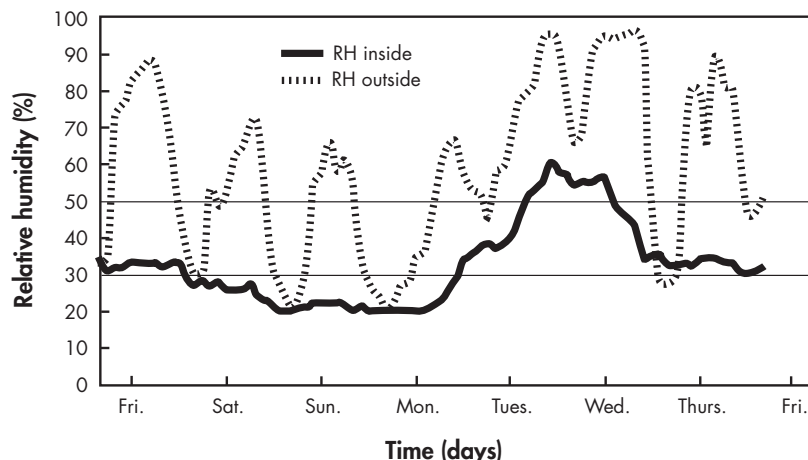
The U.S. Environmental Protection Agency (EPA) estimates that Americans spend as much as 90% of their time indoors, where levels of air pollution can be 2 to 3 times higher than outdoors.

Consequently, the EPA ranks poor IAQ among the five top environmental risks to public health. In response to this concern, a number of organizations have developed recommendations for improving residential IAQ. The American Lung Association, for example, has published *A Guide for Creating a Healthier Home*. This guide advises homeowners to

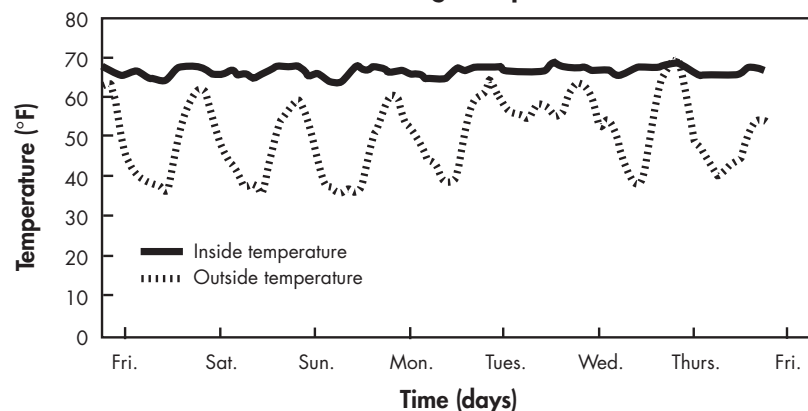
reduce indoor allergens, control moisture, avoid CO problems caused by improperly functioning combustion heating systems, select low-VOC furnishings, use household products with caution, and improve air filtration.

In most instances, healthy houses—houses that are designed to provide improved IAQ—have been developed

Second Week of Monitoring: RH



Second Week of Monitoring: Temperature



Figures 3 and 4. At the end of a second week of monitoring, with the 125–130 CFM fan with a HEPA filter running, the electric meter showed 107 kWh expended.

During the second week of monitoring (February 15–22, 2002), the supply ventilation fan with HEPA filter was turned on, and it was kept on for the entire one-week period (see Figures 3 and 4). The fan is rated at 125–130 CFM, which translates to approximately 1.17 ACH. (Only a few sizes of ventila-

tion fans with HEPA filters were available. We selected the smallest one; it was intended for a 2,000 ft² house. ASHRAE recommends a minimum of 0.35 ACH.) CO₂ levels were the same as the previous week, 380–460 ppm. VOCs were measured at the optimal level again. The electrical meter showed that

107 kWh had been expended during the previous week.

During the first week, the outdoor temperature was an average of 5°F lower than it was during the second week, and yet energy consumption was 27% higher during the second week. Outside RH was comparable for the two weeks, but during the first week, inside RH ranged from 33% to 47% (see Figure 2). The EPA and the Consumer Product Safety Commission (CPSC) recommend maintaining indoor RH levels between 30% and 50%. If humidity is too low, viral and bacterial populations tend to flourish, thus contributing to respiratory infections. If humidity is too high, fungal growth and dust mites can thrive. During the week when the ventilation fan was running, indoor RH reflected outdoor RH. Over half of the time during the second week, indoor RH was measured either above or below the 30%–50% recommended optimum range (see Figure 3).

The first two weeks of monitoring have provided preliminary evidence that running a ventilation fan 100% of the time does not improve IAQ in an unoccupied house. In fact, it contributed to indoor RH levels outside the recommended optimum range. Furthermore, running the ventilation fan led to significantly higher energy consumption by the heat pump, as well as by the fan itself. When the house is occupied, occupant activities will probably raise RH levels even further beyond the recommended range, when the ventilation fan is running and outdoor humidity is high.

On the other hand, running the fan did not lead to improved CO₂ or VOC levels. The house is not yet furnished, but it will be in the near future, and furniture is a major potential source of VOCs.

for middle- to upper-income homeowners. Floor plans are designed and materials are selected with the primary goal of improving IAQ with cost as a secondary or tertiary factor. Furthermore, when IAQ has been the first consideration, energy efficiency has often been an afterthought.

According to the U.S. Department of Health and Human Services, "low-income

and minority populations experience substantially higher rates of fatalities, hospital admissions, and emergency room visits due to asthma" than the general population. This problem is even more pronounced among children. The American Academy of Allergy and Immunology states that "asthma is 26% more prevalent among black children than

among white children." More than 10 million school days are missed annually due to asthma. Among children, asthma is the number one cause of hospitalizations. Improving residential IAQ is generally acknowledged to be one of the more effective strategies for minimizing the symptoms of this chronic disease.

Green One

The VOC device that we use is the C-11, manufactured by Eco Sensors, Incorporated, of Santa Fe, New Mexico. This device senses the level of VOC gases and vapors, such as heating gases, CO, paint thinners, dry-cleaning fluids, acetone, formaldehyde, refrigerants, ammonia, gasoline vapors, and so on. The device has a qualitative color-coded visual display

with three main levels, indicated by green, yellow, and red. Within each of these levels are three sub-levels. As an example of how the device works, a new permanent felt tip marker, which gives off emissions that you can easily smell, held 1 inch from the grille of the sensor, will produce a red display. The VOC reading in the healthy house has remained, since it was installed, at the first level of green.

CO₂ levels were also low, as would be expected in a house without occupants. It remains to be seen how much higher they will be when the house is occupied.

What's Ahead?

Monitoring and testing of the ORNL-Tuskegee healthy house will continue for the next year. During that time, research staff will run the ventilation

system intermittently to determine the effect of fresh air on the RH, IAQ, and energy consumption. Various levels of occupancy and use will be introduced in order to measure their impact on IAQ and energy consumption. During temperate seasons, windows will be opened to see what effect this has on IAQ, and to compare this with the effect of running the ventilation fan. Various strategies will be attempted to determine

how to optimize both air quality and energy efficiency.

By employing a variety of configurations, and by measuring air quality and energy efficiency, our team plans to develop guidelines for homes that balance the values of indoor air quality, affordability, and energy efficiency. As we gather more data and develop best practice strategies, we will share the results with *Home Energy* readers. Stay tuned for the next installment!



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The American Lung Association's *A Guide for Creating a Healthier Home* is available on-line as a PDF file from www.healthhouse.org/new/press.

The C-12 VOC sensor (which replaces the C-11) is available for less than \$1,000 through Eco Sensors, Incorporated
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Santa Fe, NM 87505-4737
Tel: (505)988-1421
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